

```
In [1]: #Introduction to OpenCV and Image Processing - https://www.youtube.com/watch?v=cnA-5rY3DXQ&list=PLVLQHNRLfLP9vV2\_F08f
```

```
In [2]: #pip install opencv-python
```

Requirement already satisfied: opencv-python in c:\users\sid balurkar\anaconda3\lib\site-packages (4.12.0.88)
Requirement already satisfied: numpy<2.3.0,>=2 in c:\users\sid balurkar\anaconda3\lib\site-packages (from opencv-python) (2.1.3)
Note: you may need to restart the kernel to use updated packages.

```
In [3]: import numpy as np  
import matplotlib.pyplot as plt  
%matplotlib inline
```

```
In [5]: import cv2
```

```
In [6]: img = cv2.imread(r'C:\Users\SID Balurkar\Desktop\Full Stack Data Science\Practice\labubucv.jpg')  
img
```

```
Out[6]: array([[ 59, 162, 241],
               [ 63, 166, 245],
               [ 68, 173, 252],
               ...,
               [ 49, 155, 232],
               [ 53, 161, 238],
               [ 55, 165, 241]],

               [[ 66, 169, 248],
               [ 63, 166, 245],
               [ 61, 166, 245],
               ...,
               [ 50, 154, 231],
               [ 50, 158, 235],
               [ 50, 160, 236]],

               [[ 60, 163, 242],
               [ 64, 167, 246],
               [ 64, 169, 248],
               ...,
               [ 52, 156, 233],
               [ 52, 158, 235],
               [ 49, 157, 234]],

               ...,

               [[ 62, 160, 236],
               [ 55, 155, 233],
               [ 51, 152, 230],
               ...,
               [ 53, 151, 227],
               [ 57, 158, 233],
               [ 56, 158, 233]],

               [[ 56, 154, 230],
               [ 62, 163, 238],
               [ 57, 160, 239],
               ...,
               [ 53, 154, 229],
               [ 55, 157, 232],
               [ 50, 155, 229]],
```

```
[[ 62, 160, 236],  
 [ 68, 169, 244],  
 [ 63, 166, 245],  
 ...,  
 [ 47, 156, 230],  
 [ 57, 168, 241],  
 [ 59, 172, 245]]], dtype=uint8)
```

```
In [7]: type(img)
```

```
Out[7]: numpy.ndarray
```

```
In [8]: img.shape
```

```
Out[8]: (675, 1200, 3)
```

```
In [9]: plt.imshow(img)
```

```
Out[9]: <matplotlib.image.AxesImage at 0x14f80c3d160>
```

```
In [10]: plt.show()
```



In []: *#The above image is different than the downloaded one. This is BGR. We need to convert to RGB. Open CV defaults to BGR*

```
In [23]: fix_img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
fix_img
```

```
Out[23]: array([[241, 162, 59],
               [245, 166, 63],
               [252, 173, 68],
               ...,
               [232, 155, 49],
               [238, 161, 53],
               [241, 165, 55]],

               [[248, 169, 66],
               [245, 166, 63],
               [245, 166, 61],
               ...,
               [231, 154, 50],
               [235, 158, 50],
               [236, 160, 50]],

               [[242, 163, 60],
               [246, 167, 64],
               [248, 169, 64],
               ...,
               [233, 156, 52],
               [235, 158, 52],
               [234, 157, 49]],

               ...,

               [[236, 160, 62],
               [233, 155, 55],
               [230, 152, 51],
               ...,
               [227, 151, 53],
               [233, 158, 57],
               [233, 158, 56]],

               [[230, 154, 56],
               [238, 163, 62],
               [239, 160, 57],
               ...,
               [229, 154, 53],
               [232, 157, 55],
               [229, 155, 50]],
```

```
[[236, 160, 62],  
 [244, 169, 68],  
 [245, 166, 63],  
 ...,  
 [230, 156, 47],  
 [241, 168, 57],  
 [245, 172, 59]]], dtype=uint8)
```

```
In [24]: fix_img.shape
```

```
Out[24]: (675, 1200, 3)
```

```
In [25]: plt.imshow(fix_img)  
plt.show()
```



```
In [ ]: #The above one is same as the downloaded one
```

```
In [27]: img_gray = cv2.imread(r'C:\Users\SID Balurkar\Desktop\Full Stack Data Science\Practice\labubucv.jpg', cv2.IMREAD_GRAYSCALE)  
img_gray
```

```
Out[27]: array([[174, 178, 185, ..., 166, 172, 175],
               [181, 178, 178, ..., 165, 169, 170],
               [175, 179, 181, ..., 167, 169, 168],
               ...,
               [172, 167, 164, ..., 163, 169, 169],
               [166, 174, 172, ..., 165, 168, 165],
               [172, 180, 178, ..., 166, 177, 181]], dtype=uint8)
```

```
In [28]: img_gray.shape
```

```
Out[28]: (675, 1200)
```

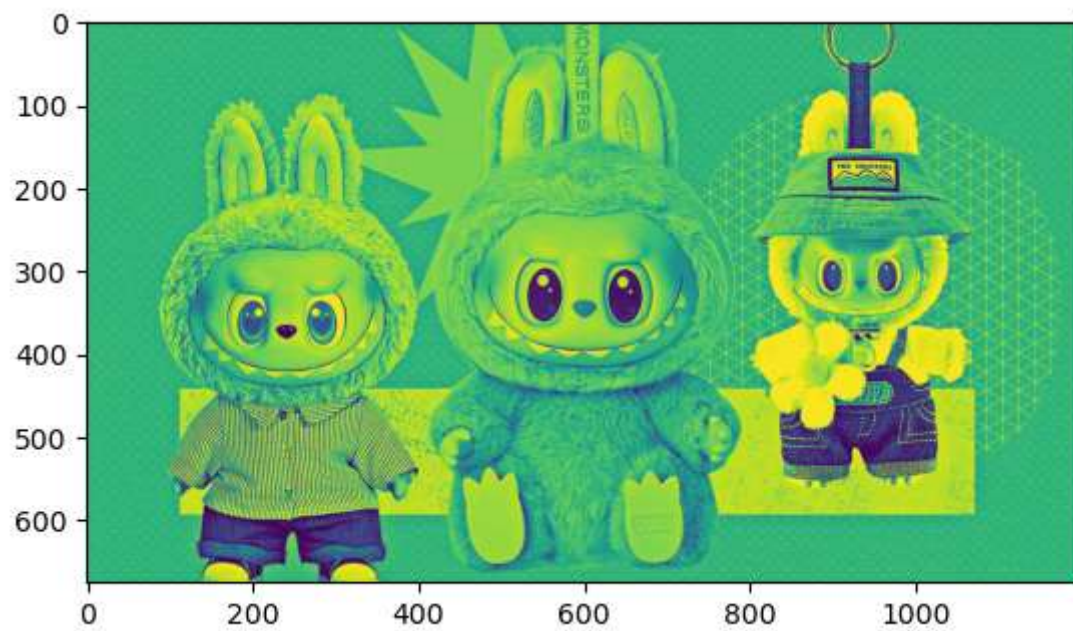
```
In [29]: img_gray.min()
```

```
Out[29]: np.uint8(0)
```

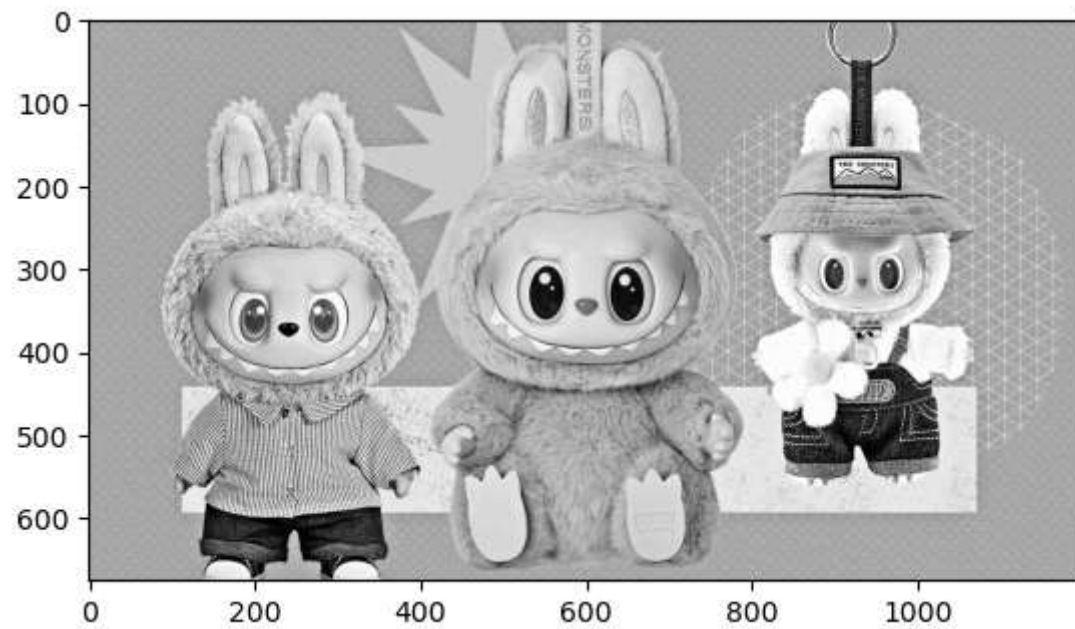
```
In [30]: img_gray.max()
```

```
Out[30]: np.uint8(255)
```

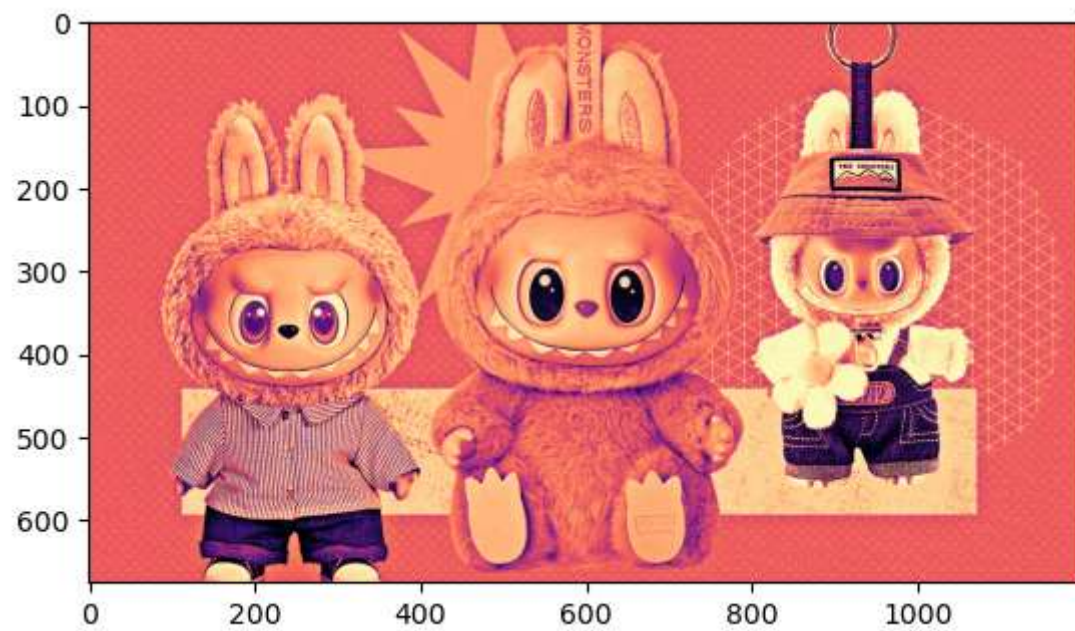
```
In [32]: plt.imshow(img_gray)
plt.show()
```



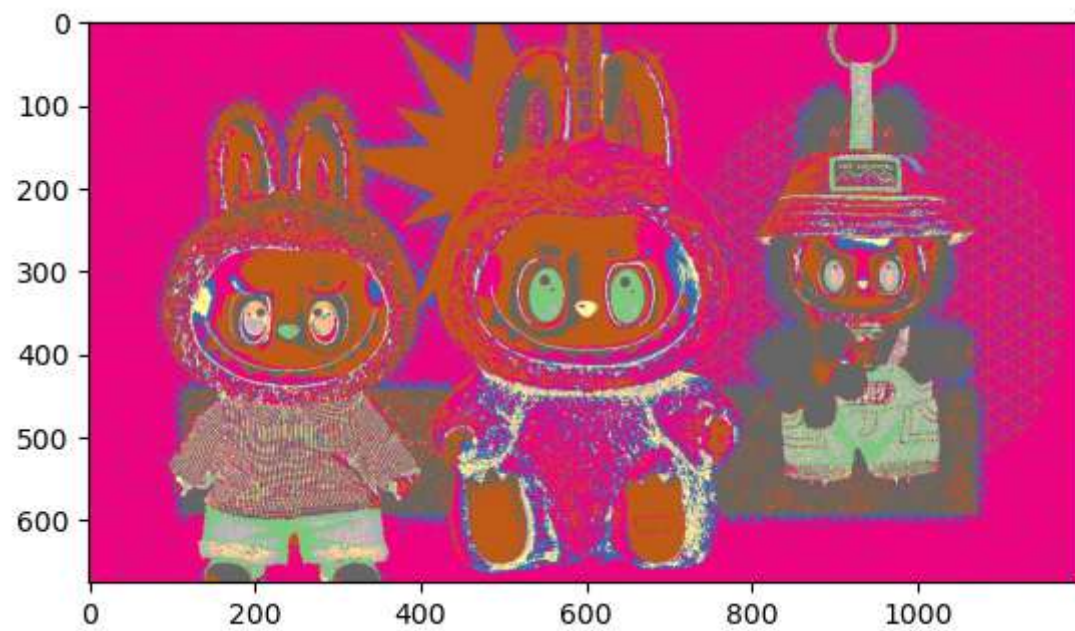
```
In [34]: plt.imshow(img_gray, cmap = 'gray')  
plt.show()
```

```
In [35]: plt.imshow(img_gray, cmap = 'magma')  
plt.show()
```



```
In [36]: plt.imshow(img_gray, cmap = 'Accent')  
plt.show()
```



```
In [37]: plt.imshow(fix_img)
plt.show()
```



```
In [38]: fix_img.shape
```

```
Out[38]: (675, 1200, 3)
```

```
In [41]: fix_img_1 = cv2.resize(fix_img,(337,600))  
fix_img_1
```

```
Out[41]: array([[247, 168, 64],
               [240, 162, 54],
               [242, 163, 58],
               ...,
               [229, 150, 56],
               [239, 160, 66],
               [236, 159, 52]],

               [[245, 166, 63],
               [245, 167, 59],
               [235, 156, 51],
               ...,
               [225, 146, 52],
               [238, 159, 63],
               [234, 157, 50]],

               [[246, 167, 63],
               [238, 160, 52],
               [236, 157, 52],
               ...,
               [224, 146, 49],
               [235, 156, 60],
               [235, 158, 54]],

               ...,

               [[231, 153, 54],
               [233, 155, 48],
               [228, 151, 46],
               ...,
               [230, 153, 50],
               [233, 156, 58],
               [231, 155, 54]],

               [[237, 161, 59],
               [234, 154, 47],
               [232, 156, 51],
               ...,
               [228, 155, 51],
               [234, 160, 63],
               [231, 156, 54]],
```

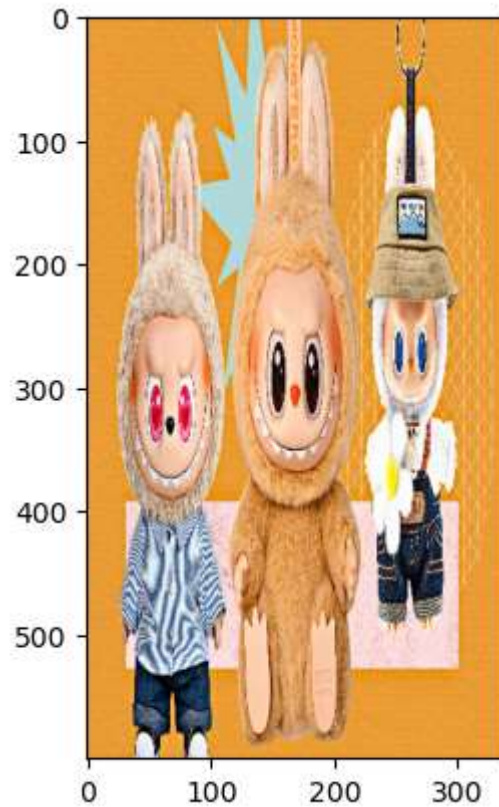
```
[[244, 168, 66],  
 [241, 160, 53],  
 [233, 157, 52],  
 ...,  
 [227, 154, 48],  
 [227, 154, 52],  
 [237, 164, 54]]], dtype=uint8)
```

```
In [42]: fix_img_1.shape
```

```
Out[42]: (600, 337, 3)
```

```
In [ ]: #The above size is changed to 50% of the actual image
```

```
In [43]: plt.imshow(fix_img_1)  
plt.show()
```



```
In [45]: w_ratio = 0.5  
         h_ratio = 0.5  
         fix_img_2 = cv2.resize(fix_img,(0,0),fix_img,w_ratio,h_ratio)
```

```
In [46]: fix_img_2
```

```
Out[46]: array([[245, 166, 63],
               [250, 171, 66],
               [246, 168, 60],
               ...,
               [237, 158, 62],
               [231, 153, 50],
               [238, 161, 52]],

               [[241, 162, 59],
               [245, 166, 61],
               [237, 159, 51],
               ...,
               [234, 156, 58],
               [235, 157, 56],
               [236, 159, 54]],

               [[233, 154, 51],
               [232, 153, 48],
               [233, 155, 47],
               ...,
               [234, 156, 56],
               [237, 159, 59],
               [234, 156, 56]],

               ...,

               [[229, 151, 53],
               [230, 152, 52],
               [234, 157, 53],
               ...,
               [234, 153, 52],
               [233, 154, 52],
               [228, 153, 49]],

               [[234, 158, 59],
               [235, 156, 52],
               [235, 156, 48],
               ...,
               [235, 160, 63],
               [228, 153, 54],
               [232, 157, 55]],
```



```

[[240, 164, 65],
 [243, 164, 59],
 [242, 162, 54],
 ...,
 [227, 154, 51],
 [226, 152, 46],
 [243, 170, 58]]], dtype=uint8)

```

```

In [47]: plt.imshow(fix_img_2)
plt.show()

```



```

In [ ]: #Observe that the shapes axis are reduced by 50%

```

```

In [48]: img3 = cv2.flip(fix_img_2,0)

```

```

In [49]: plt.imshow(img3)
plt.show()

```



In []: *#The above is flipped*

```
In [50]: img4 = cv2.flip(fix_img_2,1)
plt.imshow(img4)
plt.show()
```



In []: *#Observe that the right and left Labubu are swapped, means this is an mirror image*

```
In [51]: img5 = cv2.flip(fix_img_2,2)
plt.imshow(img5)
plt.show()
```



```
In [ ]: #The above is the same one now as previous one img4 and ing3 are same
```

```
In [54]: img5 = cv2.flip(fix_img_2,3)
plt.imshow(img5)
plt.show()
```



```
In [ ]: #Lets save the img4 image
```

```
In [55]: cv2.imwrite('newflippedimagelabubu.jpg',img4)
```

```
Out[55]: True
```

```
In [56]: pwd
```

```
Out[56]: 'C:\\Users\\SID Balurkar\\VSCODE_PROJECT'
```

```
In [ ]:
```